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0 RETURN
0 REM CALC SIDEREAL TIME
0 GC = 11.927485
0 TC = .065711
0 T2 = TC * (M1 - 7020) + GC
0 IF T2 > 24 THEN T2 = T2 - 24: GOTO 1660
0 IF T2 < - 24 THEN T2 = T2 + 24: GOTO 1670
0 IF T2 < 0 THEN T2 = T2 + 24
0 RETURN
0 REM GET TIME DIFFERENCES
0 A1 = SIN ( FN RAD(V(3))) + SIN ( FN RAD(LA))
0 IF A1 < 0 THEN A1 = ABS (A1): GOTO 1740
0 IF A1 > 0 THEN A1 = - A1
0 A1 = (A1 - .01454) / COS ( FN RAD(LA))
0 A1 = A1 / COS ( FN RAD(V(3)))
0 B1 = SIN ( FN RAD(D(1))) + SIN ( FN RAD(LA))
0 IF B1 < 0 THEN B1 = ABS (B1): GOTO 1790
0 IF B1 > 0 THEN B1 = - B1
0 B1 = (B1 - .00989) / COS ( FN RAD(LA))
0 B1 = B1 / COS ( FN RAD(V(1)))
0 C1 = 4 * (A1 - B1)
0 D1 = 60 * (R(3) - R(1))
0 M1 = C1 + D1
0 REM ADJUST FOR RA PASSING ZERO
0 IF M1 > 720 THEN M1 = M1 - 1440: GOTO 1870
0 IF M1 < - 720 THEN M1 = M1 + 1440
0 IF M1 < 0 THEN ASS = "AN EVENING STAR ":MSS = "SETTING":BAS = " AFTER"
0 IF M1 > 0 THEN ASS = " A MORNING STAR ":MSS = "RISING ":BAS = " BEFORE"
0 M1 = ABS (M1)
0 M1$ = LEFT$ ( STR$ (M1),4)
0 PRINT P$ (I); " IS ";ASS
0 PRINT " ";MSS; " ";M1$; " MINUTES ";BAS; " THE SUN"
0 PRINT : PRINT : PRINT
0 RETURN

```